

SMART INDIA HACKATHON 2026

BitForensics

AI-Powered Monitoring of Bitcoin Transaction Traffic

Problem Statement 26146

National Technical Research Organisation (NTRO)

How the project works · Models · Tech flow

Simple notes for a live presentation

Offline · Linux / Windows / macOS · No live Bitcoin node required

1. One-minute story (say this first)

Bitcoin wallets have no name. Criminals can move ransomware money, darknet proceeds, or extortion payments by splitting funds across many addresses. Banks cannot see this the usual way. NTRO asked for an **offline** tool that reads a bulk dump of Bitcoin metadata, joins **network clues** (IP, port, time) with **on-chain clues** (TXID, wallets, amount), and uses a **real ML model** — not only if-else rules — to rank suspicious activity with a reason and a confidence score.

Bolne ke liye: “Yeh software dump ko locally padhta hai, graph banata hai, Isolation Forest se outliers nikalta hai, aur investigator ko ranked leads deta hai.”

2. User kya karta hai (product flow)

- **Start:** npm run dev ya python run.py → **http://localhost:8000**.
- **Upload** CSV / JSON / XML (demo: ~1 lakh rows wali CSV).
- Overview par raho. **Run Analysis** dabao (upload se model khud train nahi hota).
- Stats + poori alert list dekho. Row click → TXID, IP, geo, AML flags.
- **Link Graph:** neela = IP, hara = wallet, narangi = transaction, laal border = flagged.
- Optional: GeoIP ek baar download; uske baad fully offline.

3. Tech stack

Layer	Kya use kiya, kyun
Language	Python 3.9+ — ingest, ML, API ek hi language.
Server	FastAPI + Uvicorn — JSON API + dashboard files.
Database	SQLite file. Cloud nahi. Offline.
ML	scikit-learn. GPU / paid API nahi. Train isi dump par.
Anomaly	Isolation Forest — unsupervised outliers.
Cluster	MiniBatchKMeans — 1 lakh rows laptop par.
Scale	StandardScaler — amount aur hour ek scale par.
GeoIP	CSV fallback, ya DB-IP Lite MMDB (CC BY 4.0).
UI	HTML / CSS / JS — npm se frontend build nahi.
OS	Linux (PS), Windows, macOS.

4. End-to-end tech flow

Step	Code	Kaam
1. Ingest	ingest/parser.py	CSV/JSON/XML → ek format: time, IP, port, TXID, wallets, amounts, fee, script, country, ASN.
2. Store	SQLite	Unique TXID. Do files mein same TXID = ek row. Missing amount/IP doosri file se fill.
3. Geo	geo/lookup.py	Country/ASN na ho to IP se offline lookup (MMDB ya CSV).

4. Correlate	correlation/engine.py	Same TXID + IP. Score 0–1: IP coverage, repeat IP, wallets, amount. Window 5 min.
5. Graph	graph/builder.py	IP ↔ transaction ↔ wallet. Flagged = red border.
6. Features	ml/analyzer.py	Har tx → 11 numbers (log amount, fee ratio, in/out, hour, geo, ...).
7. ML	Forest + KMeans	Isi dump par train. Outliers + similar groups.
8. Explain	typology + red_flags	Flag ke baad label (mixer/peel/...) aur 120 AML conditions.
9. Rank	/api/alerts	Anomaly + cluster + strong correlation. Score se sort. Confidence %.
10. UI	frontend/dist	Overview, Alerts, Graph, Transactions, Clusters, Upload.

Bolne ke liye: "Parse → SQLite → IP-chain join → 11 features → Isolation Forest → cluster → explainable list."

5. Models kaise banaye

5.1 Isolation Forest (mukhya model)

Yeh **unsupervised** hai. Hum “criminal / clean” label nahi dete (seized live data nahi milta). Forest feature space ko random splits se kaatta hai. Jo point jaldi alag ho jata hai, woh **outlier** hai. Yeh “ajeeb mixer / whale / zyada outputs” ke liye ek hi threshold rule se better hai.

Train kab: sirf **Run Analysis** par. Current SQLite rows par fit, phir `data/models/isolation_forest.joblib` save.

Gyarah features

- **log_amount, log_fee** — chhoti aur badi BTC values ek model mein.
- **fee_ratio** — $\text{fee} \div \text{amount}$.
- **input_count, output_count, io_ratio** — kitne wallets se aaya / kitne ko gaya (mixer peel).
- **has_ip** — network observation mili ya nahi.
- **hour** — raat ko burst.
- **is_p2sh** — complex script type.
- **foreign_geo** — desh local/unknown nahi.
- **amount_z** — is dump ke average se kitna door (z-score).

Settings (hyperparameters)

Cheez	Value / matlab
Library	sklearn IsolationForest
Scaler	StandardScaler pehle, phir forest
random_state	42 — same file, same demo
contamination	≈6% (n≥50k), 7% (n≥10k), 10% (chhoti file). Ye outlier share hai, 46 alerts ki cap nahi.
n_estimators	100–150 trees
max_samples	max 12,000 subsample — 1 lakh rows tez
n_jobs	-1 (saare CPU cores)
Score	-decision_function (zyada = zyada isolated)
Confidence	us score ka percentile, 8–99%
Severity	CRITICAL / HIGH / MEDIUM
Filter	≈0 BTC rows skip — khali amount ko mixer mat banao

Sawal “ML hai ya rules?": “Forest decide karta hai kaun flagged hai. Rules sirf explanation likhte hain.”

5.2 MiniBatchKMeans (clustering)

PS clustering maangta hai. 1 lakh points par DBSCAN laptop par slow ho sakta hai, isliye **MiniBatchKMeans**. Wallets (agar address hon) ya txs ko volume, counts, linked IPs se group karta hai. Har group ek cluster + kabhi HIGH lead (jaise 12 linked mixer-like txs).

Cheez	Value
Library	sklearn MiniBatchKMeans

n_init	3
batch_size	1024 (wallets) / 2048 (tx sample)
random_state	42

5.3 Model ke baad (explainability)

- **Typology:** mixer, peel-chain, consolidation, whale, dusting, structuring, pass-through, overnight — alert ka title.
- **120 AML/KYT:** jo metadata allow kare (fan-out, geo, time). Mixer-list / sanctions / darknet **apne aap claim nahi**. Disclaimer: indicator, crime ka proof nahi.

6. Network ↔ blockchain correlation

Agar row mein src/dst IP hai, woh TXID ki network observation hai. TXID se group, wallets + IPs collect, score: dono IP, IP kitni baar, wallet count, amount. Window **300 second**. Bina IP ke dump par khali correlation skip — 1 lakh CSV tez rehti hai. Graph wahi join ka picture hai.

7. Alerts kaise rank hoti hain

- Har Isolation Forest hit (asli amount) = ek lead.
- Har mota cluster = ek lead.
- Kuch strong IP–TX correlations = medium leads.
- Alerts page **saari** list (100k par ~6,000), fake 40 nahi. Row click = evidence.

8. Demo mein kya dikhana hai

Page	Jury ko dikhao
Overview	Unique txs, anomalies, alerts, BTC volume, poori table.
Link Graph	IP / wallet / tx; laal = flagged.
Alerts	HIGH row → TXID, explorer, IP, country/ASN, why, 120 flags.
Transactions	Raw fields.
Clusters	KMeans groups.
Upload	CSV/JSON/XML, GeoIP, Clear Data.

9. 3-minute bolna (script)

- “PS 26146: NTRO ke liye offline Bitcoin forensic system.”
- localhost:8000. “Koi cloud API nahi. SQLite disk par.”
- CSV upload. “Parser CSV, JSON, XML — column names flexible.”
- Run Analysis. “Ab Isolation Forest is dump par train hota hai; KMeans cluster karta hai.”
- Alert count. “Yeh model ka outlier rate hai, hard-coded 46 nahi.”
- Ek alert kholo. “Confidence forest se. Typology shape se. 120-list explanation hai, court verdict nahi.”
- Graph. “IP TXID ke through wallets se juda.”
- Linux: “Haan. Windows: Git + Python PATH + Node, phir npm run dev.”

10. Run kaise karein

Python 3.9+ aur Node.js. `git clone`

`https://github.com/ashutoshmishra52/bitforensics.git → cd bitforensics → npm install → npm run dev → http://localhost:8000.`

Sirf Python: `venv banao, pip install -r requirements.txt, python run.py.`

11. Limits (agar poochhein)

- Synthetic/demo data — real intercept nahi.
- Unsupervised: unusual ≠ guilty. Leads hain, conviction nahi.

- Bade dump par graph sample hota hai.
- Sanctions/mixer blacklist alag intel se aati hai; CSV mein pretend nahi.

GitHub: <https://github.com/ashutoshmishra52/bitforensics> · Short write-up: APPROACH.md